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# Bronchiolitis

## Introduction

Bronchiolitis is an acute inflammatory condition of the bronchioles that is a result of virus-induced injury.

## Etiology

Respiratory syncytial virus (RSV) is the most common viral agent isolated in about 75% (30–70% in Indian studies).

*Other viruses*: Rhinovirus, parainfluenza, adenovirus, human metapneumovirus, and bocavirus are the other viruses commonly causing the condition. Mycoplasma is more frequently implicated in older children with bronchiolitis.

## Diagnosis

- ☑ Persistent cough, following a prodrome of coryza lasting 1–3 days, with tachypnea with or without chest recessions and wheeze and/or crackles occurring in a child <2 years of age (usually below 1 year of age, with a peak between 3 and 6 months).
- ☑ Associated fever, usually below 39°C, in around 30% cases and poor feeding, vomiting usually after 3–5 days of illness.
- ☑ Apnea may be the only presenting feature, particularly below 6 weeks of age.
- ☑ The chest may appear hyperexpanded and may be hyper-resonant to percussion. Wheezes and fine crackles may be heard throughout the lungs.

## Indications for Hospitalization

- ☑ Persistent tachypnea >60 breaths/minute or respiratory distress in form of grunting, recessions
- ☑ Inadequate oral intake, inability to feed, dehydration, and inadequate fluid intake (50–75% of usual volume)
- ☑ Oxygen saturation ($SpO_2$) <92% in room air
- ☑ Child appears seriously unwell to the healthcare provider
- ☑ Skill and confidence of the caregiver to look after the child at home and distance from the hospital

## Signs of Severe Bronchiolitis

- ☑ Apnea, observed/reported
- ☑ Marked respiratory distress (severe grunting/chest indrawing/tachypnea >70/minute)
- ☑ Central cyanosis, or $SpO_2$ below 90% (age > 6 weeks) or below 92% (age < 6 weeks, or any age with underlying health conditions)

## Risk Factors for Severe Bronchiolitis

Predictors of severe bronchiolitis are presented in **Table 1**.

**TABLE 1:** Predictors of severe bronchiolitis.

| *A. Host-related risk factors* | *B. Environmental risk factors* | *C. Clinical predictors* |
|---|---|---|
| ☑ Prematurity, especially <32 weeks of gestation<br>☑ Low birth weight<br>☑ Age <6–12 weeks<br>☑ Chronic lung disease including BPD<br>☑ Hemodynamically significant congenital heart disease (e.g., moderate-to-severe pulmonary hypertension, cyanotic heart disease, or congenital heart disease that requires medication to control heart failure)<br>☑ Immunodeficiency<br>☑ Neuromuscular disorders | ☑ Having older siblings<br>☑ Passive smoke<br>☑ Household crowding<br>☑ Child care attendance<br>☑ Lower socioeconomic status | ☑ Toxic or ill appearance<br>☑ Oxygen saturation <95% by pulse oximetry while breathing room air<br>☑ Respiratory rate 70 breaths per minute<br>☑ Moderate/severe chest retractions<br>☑ Atelectasis on chest radiograph |

## Differential Diagnosis

- ☑ *Pneumonia*: Fever >39°C with persistent focal crackles
- ☑ *Episodic viral wheeze*: Persistent wheeze without crackles, or recurrent episodes with or without a family history of atopy

## Investigations

- ☑ Is a clinical diagnosis based on age, seasonal occurrence, typical clinical presentation, and physical examination?
- ☑ Blood investigations and radiology is routinely not indicated.
- ☑ A pulse oximetry reading helps to identify hypoxia and need for admission.
- ☑ Investigations in admitted patients to rule out alternate diagnosis such as bacterial pneumonia, congenital heart disease with failure, or sepsis might occasionally be indicated.
- ☑ Admitted babies may need an arterial blood gas (ABG) analysis, complete blood count, C-reactive protein (CRP), serum electrolytes, and chest radiography for managing the more serious patients.
- ☑ Measurement of lactate dehydrogenase (LDH) concentration in the nasal-wash fluid has been proposed as an objective indicator of bronchiolitis severity **(Table 2)**.
- ☑ Identification of viral agents does not affect management in the majority of patients.

However, in the hospital setting, to avoid antibiotic abuse and prevent nosocomial transmission may be done by:

- ☑ Antigen detection, immunofluorescence, polymerase chain reaction (PCR), and culture of respiratory secretions obtained by nasal wash or nasal aspirate.
- ☑ New techniques such as real-time PCR, nested PCR, and multiplex PCR have improved the virologic diagnosis of bronchiolitis immensely.

**TABLE 2:** Severity of bronchiolitis.

| | ***Mild*** | ***Moderate*** | ***Severe*** |
|---|---|---|---|
| Feeding ability | Normal ability to feed | Appear short of breath during feeding | May be reluctant or unable to feed |
| Respiratory distress | Little or no respiratory distress | Moderate distress with some chest wall retractions and nasal flaring | ☑ Severe distress with marked chest wall retractions, nasal flaring and grunting<br>☑ Can have frequent and prolonged apnea |
| Saturation | Saturation >92% | Saturation <92%, correctable with $O_2$ | Saturation <92%, may or may not be correctable with $O_2$ |

Management

- ☑ Treatment is focused on symptomatic relief and maintaining hydration and oxygenation.
- ☑ Fever should be controlled with paracetamol.
- ☑ Nose block should be cleared with saline nasal drops and gentle suctioning.
- ☑ Child should be made to lie in a propped up or head end elevated positioning.
- ☑ Orogastric tube feeding may be indicated in admitted patients. Intravenous (IV) fluids in children with impending respiratory failure or who do not tolerate orogastric/nasogastric (OG/NG) fluids.
- ☑ Suctioning of the upper airway in children with apnea, respiratory secretions, and feeding difficulties due to upper airway secretions
- ☑ Supplemental oxygen in children with $SpO_2$ below 90% (>6 weeks) or below 92% (<6 weeks or with underlying health issues)
- ☑ Continuous positive airway pressure (CPAP) in babies with impending respiratory failure (limited low-quality evidence)
- ☑ High-flow nasal cannula (HFNC) oxygen may have a role as a rescue therapy to reduce proportion of those requiring intensive care
- ☑ Drugs with questionable value might reduce need for admission or length of hospital stay, but broad consensus is lacking.
  - *Nebulized hypertonic saline*: In children hospitalized for >3 days
  - *Nebulized adrenaline*: 0.1–0.3 mL/kg/dose of 1:1,000 as a potential rescue medication; however inconsistent and short-lived improvement
  - *Beta-agonists*: Optional single trial; may be continued if there is clinical response (a trial of bronchodilator therapy may be initiated, but should be discontinued if there is no objective improvement)
- ☑ No role of:
  - Chest physiotherapy
  - Antibiotics
  - Antivirals
  - Montelukast
  - Ipratropium bromide
  - Systemic or inhaled steroids
  - Steam inhalation
  - RSV polyclonal immunoglobulin/palivizumab (no roll in acute management but useful in prophylaxis)
  - Inhaled furosemide/inhaled interferon alfa-2a/inhaled recombinant human deoxyribonuclease (DNase)
- ☑ Interventions which are possibly effective for most severe cases:
  - CPAP
  - Surfactant
  - Heliox
  - Aerosolized ribavirin

## Criteria for Discharge

- ☑ Clinically stable
- ☑ Taking adequate oral feeds, at least 75% of usual
- ☑ Maintaining $SpO_2$ above 90% (>6 weeks) and 92% (<6 weeks or with health issues) in room air
- ☑ Ability of the caregiver to look after at home and distance from the hospital, and have understood the "red flag" signs

Red flag signs for the caregiver at home:

- ☑ Increased work of breathing (e.g., grunting, nasal flaring, and chest retractions)
- ☑ Fluid intake <50–75% of normal or no urine for 12 hours
- ☑ Apnea or cyanosis
- ☑ Exhaustion (i.e., not responding normally to social cues and responds only with prolonged stimulation)

## Prevention

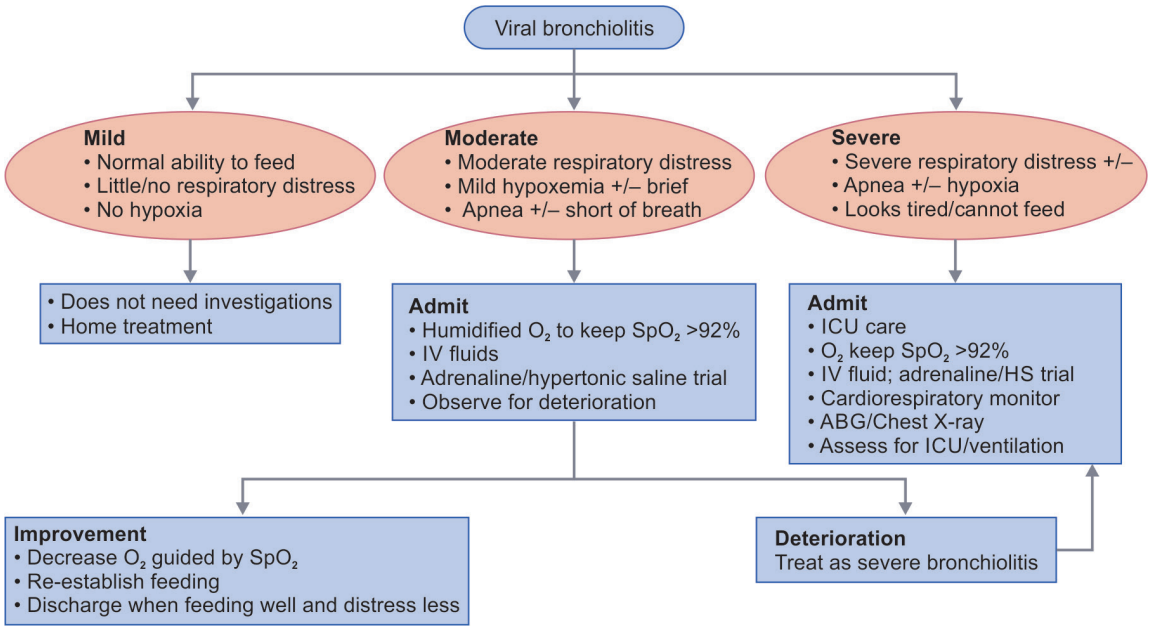
- ☑ *Breastfeeding*: Three-fold greater risk in non-breastfed infant
- ☑ Hand hygiene
- ☑ Avoid passive smoking
- ☑ *Immune prophylaxis:*
  - *Palivizumab*: Monoclonal antibody, monthly injections during seasonal epidemics
    - *Indications*: Infants <12 months with prematurity <29 weeks; CLD of prematurity; hemodynamically significant heart disease
    - Palivizumab is administered intramuscularly at a dose of 15 mg/kg monthly (every 30 days) during the RSV season. A maximum of five doses is generally sufficient prophylaxis during one season.
  - *Nirsevimab*: On trial; single dose for 5 months
  - *Motavizumab*, a second-generation mAb, and Numax-YTE, a third-generation mAb—under trial

## Complications

- ☑ Acute respiratory distress syndrome (ARDS)
- ☑ Myocarditis
- ☑ Congestive heart failure
- ☑ Arrhythmias
- ☑ Bronchiolitis obliterans
- ☑ Secondary bacterial infection
- ☑ Predisposition to childhood asthma

## Summary

**Flowchart 1:** Summary of viral bronchiolitis.




(ABG: arterial blood gas; ICU: intensive care unit; IV: intravenous; $SpO_2$: oxygen saturation)

## Further Reading


- ☑ Gupta S, Shamsundar R, Shet A, Chawan R, Srinivasa H. Prevalence of respiratory syncytial virus infection among hospitalized children presenting with acute lower respiratory tract infections. Indian J Pediatr. 2011;78:1495-7.
- ☑ National Institute for Health and Care Excellence. (2015). Bronchiolitis in children: diagnosis and management. [online] Available from www.nice.org.uk/guidance/ng9. [Last accessed March, 2022].
- ☑ Ralston SL, Lieberthal AS, Meissner HC, Alverson BK, Baley JE, Gadomski AM, et al. Clinical practice guideline: the diagnosis, management and prevention of bronchiolitis. Pediatrics. 2014;134(5): e1474-502.

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